Meet Babariya

AI Engineer · ML on technical & sensor data · LLM systems and evaluation

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Summary

Business Information Systems student (grade 1.55) who builds and validates AI systems end to end. I built an **EV-motor virtual temperature sensor that beats the published benchmark**, with an evaluated LLM assistant deployed as a serverless API. At **Datacurve (Y Combinator)** I design ML benchmark challenges, many on real sensor data, used to train and test AI agents. Kaggle Competitions Expert (top 1%). Seeking an **AI / ML engineering internship** in Germany.

Experience

Datacurve (Y Combinator) — Project Eris Remote

ML Benchmark Engineer (Contract) *Sep 2026 – Present*

- Design end-to-end ML benchmark challenges used to train and evaluate AI coding agents: dataset curation, split pipeline (*prepare.py*), evaluator (*grade.py*), scoring rubrics and reference solution notebooks.
- Built 25+ challenge designs across **vision fine-tuning, sequence-to-sequence, from-scratch and LLM-evaluation** tracks, many on **real multi-channel sensor and telemetry data** (air-quality stations, ocean buoys, solar irradiance, wastewater plants), e.g. reconstructing a withheld sensor channel from the others.
- Calibrated difficulty so frontier agents score below 0.8 (baseline ladders up to GPU fine-tunes, exact Bayes-optimal decoders to prove information ceilings) and hardened graders against label leakage and test-set lookup.

Jainam Broking Surat, India

Derivatives Trader *Apr 2022 – Dec 2024*

- Traded weekly NIFTY / BANKNIFTY options with delta-neutral strategies and real-time, Greeks-based hedging.

Education

Furtwangen University (HFU) Furtwangen, Germany

B.Sc. International Business Information Systems *Mar 2025 – Present*

- Current grade: **1.55** · Transcript

Projects

EV-Motor Virtual Temperature Sensor + LLM Engineering Assistant | Case study & live demo | GitHub

Python, PyTorch, scikit-learn, Gemini API, Vercel, Streamlit

- Estimates rotor-magnet and stator temperatures of a PMSM from vehicle-measurable signals (185 h test-bench data). Ensemble of a load-history linear model and a physics-structured **thermal neural network**: **1.50 K² MSE / 5.73 K worst case** on the published benchmark (best published: 2.87 / 6.02), without torque as an input.
- Found that the benchmark's test cycles contain only cold starts. After a hot restart a cold-start assumption is ~30 K wrong for 40 min, which I fixed with stored-state initialisation. All choices were made by grouped cross-validation.
- LLM assistant (hand-written agent loop, 6 deterministic tools, fallback across free models) deployed as a **serverless API**. It scores **13/13** on a graded evaluation in production; the evaluation caught 3 real bugs.

QuantPilot — AI copilot & quant-ranked copy trading | *TypeScript, Next.js, Python, FastAPI, PostgreSQL* *code private*

- Full-stack product on Hyperliquid: TypeScript execution API and bot over a Python service that ranks traders by risk-adjusted returns with a Deflated-Sharpe filter (run on live mainnet data).

LLM tools for finance | RAG · Agent | *Claude API, ChromaDB, tool use*

- Cited question answering over SEC filings (local embeddings, vector search, grounded prompt), and a tool-using analyst agent that chains VaR / volatility / Sharpe tools.

German Traffic Sign Recognition | GitHub | *PyTorch CNN, 43 classes (ADAS perception),* ***97.7% test accuracy***

Competitions & Achievements

- **Kaggle Competitions Expert** (top 1% globally): Silver, CIBMTR (120th / 3,325); Bronze, MITSUI Commodity Prediction (165th / 1,711).
- **IMC Prosperity 4**: 230th / 18,803 teams, **12th in Germany** · **WorldQuant BRAIN** Gold · AlgoTrade Hackathon 2025 · WQU Applied Data Science Lab.

Technical Skills

LLM / GenAI	Claude, Gemini & OpenAI APIs, agents & tool use, RAG, embeddings, vector DBs (ChromaDB), LLM evaluation, benchmark & rubric design
ML / Deep Learning	PyTorch, scikit-learn, time-series & sensor-data feature engineering, physics-structured / state-space models, grouped cross-validation, Hugging Face, CNNs
Software	Python, JavaScript / TypeScript, SQL; FastAPI, Streamlit, Vercel serverless, Next.js / Astro, PostgreSQL, Docker, Git; pandas, NumPy, Power BI
Languages	English (fluent), German (A2, actively learning)